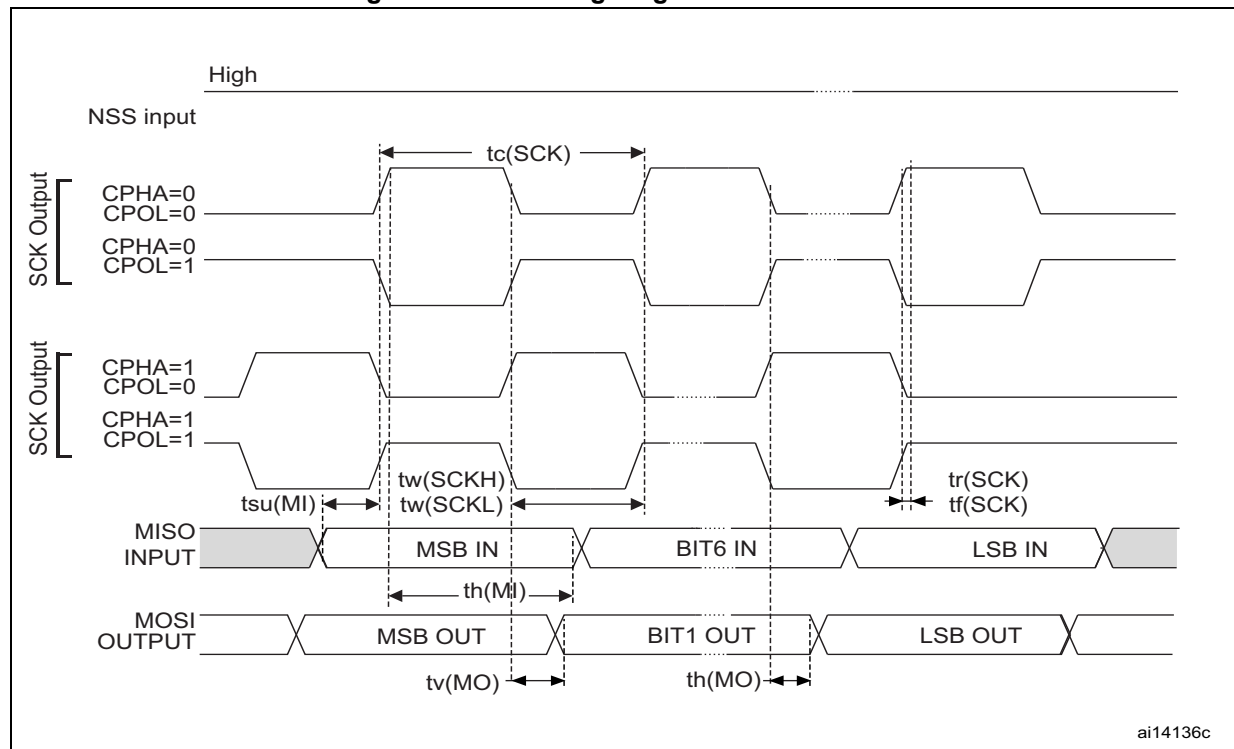


Figure 31. SPI timing diagram - master mode



1. Measurement points are done at 0.5 V_{DD} and with external C_L = 30 pF.

Table 65. I²S characteristics

Symbol	Parameter ⁽¹⁾	Conditions	Min	Max	Unit
f _{MCK}	I2S main clock output	-	-	48	MHz
f _{CK}	I2S clock frequency	Master TX	-	12	MHz
		Master RX	-	12	
		Slave TX	-	15	
		Slave RX	-	48	
t _{v(WS)}	WS valid time	Master mode	-	5	ns
t _{h(WS)}	WS hold time	Master mode	0	-	ns
t _{su(WS)}	WS setup time	Slave mode	3.5	-	ns
t _{h(WS)}	WS hold time	Slave mode	1	-	ns
t _{su(SD_MR)}	Data input setup time	Master receiver	5	-	ns
t _{su(SD_SR)}		Slave receiver	2.5	-	ns
t _{h(SD_MR)}	Data input hold time	Master receiver	2.5	-	ns
t _{h(SD_SR)}		Slave receiver	1	-	ns
t _{v(SD_ST)}	Data output valid time	Slave transmitter (after enable edge)	-	19.5	ns
t _{v(SD_MT)}		Master transmitter (after enable edge)	-	5	ns